

Data Science

NLP Pipelines

NLP and GenAI



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Define an NLP Pipeline and Hugging Face Pipeline API.

2

Explain the framework for selecting a pre-trained model.

3

Construct a pipeline for sentiment analysis using encoder models and compare performance across models.

NLP Pipelines

Evolution of traditional and modern NLP

TRADITIONAL NLP

Requires training a custom model for your data.

Requires large amounts of labeled data.

Not typically generalizable to new use cases.

MODERN NLP

Uses a pre-trained model.

Requires less training data.

Faster and more scalable.

More generalizable.

You will focus on Modern NLP methods in this lesson.

What is a pipeline?

A **pipeline is a standardized set of steps** that process raw text or images, perform model inference, and delivers model outputs.

Text for humans

I'm learning NLP



[0, 1, 0, 0, 1, 0]
[1, 0, 0, 0, 1, 0]
[0, 0, 0, 1, 0, 1]

A pipeline helps across the whole NLP General Workflow

Raw Text

Data Cleaning

Pre-processing
for computers

Model

Application

We start with a collection of raw text.

This group of documents is your **corpus**.

A **document** is a piece of text that represents one unit of analysis from the corpus.

Steps vary by corpus. Common actions include:

- **Removing duplicates** to speed up models and reduce cost.
- **Removing identifying information** such as names for privacy.
- **Removing stop words** (common words like **a**, **the**, etc.).
- Simplifying text using methods such as **stemming/lemmatization**.

Tokens or **N-grams** are the unit of analysis. Often these are **words or pieces of words** or even characters. You must **use the same tokenization method your model used for training** (BERT Tokenizer → BERT model). Units are then **converted into numbers** the computer can use.

Typically, you will use a **pre-trained model** to that you apply to your text. These models are built vast amounts of training data (think all of Facebook, all of GitHub, all books, all of reddit, all scientific journal articles, the full web). You can also **build a model from scratch** but this **is very, very rare**.

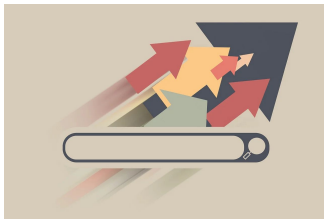
This is where you use your text for some purpose such as **summarization, translation, classification**.



Common Use Cases

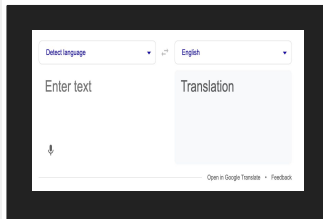
1

Searching the
web



2

Translation



3

Summarization
of documents



4

Identifying
entities

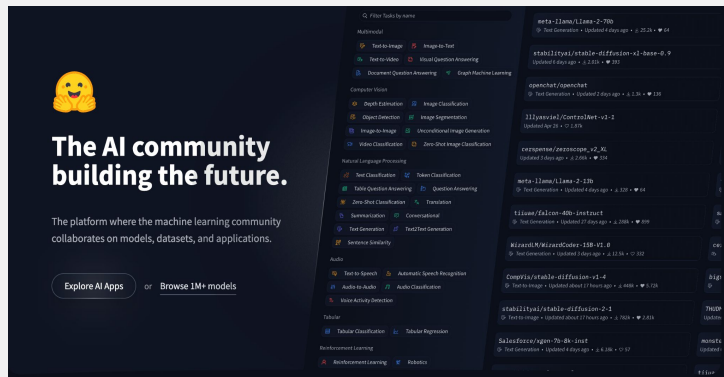


Hugging Face Pipelines

Hugging Face

Hugging Face is a platform that hosts pre-trained models and provides tools that help you use pre-trained models on your data.

One of those tools is the **Pipeline API** which helps you process your data and use it in a pre-trained model with a lot less code than would be required if you did it by hand.



Pipeline API

The **Hugging Face Pipeline API** is a wrapper that abstracts many of the NLP workflow steps needed for transformer NLP models, which take text input and generate some output, into a few lines of code.

It is a wrapper for all other Hugging Face pipelines.



Pipeline API

It is a wrapper for all other Hugging Face pipelines.

From this:

```
# Initialize a CountVectorizer and predict star ratings
vectorizer = CountVectorizer(ngram_range=(1, 2), stop_words='english')
# Fit and transform the training data
X_train_transformed = vectorizer.fit_transform(X_train)
# Transform the testing data
X_test_transformed = vectorizer.transform(X_test)
# Initialize a Logistic Regression model
model = LogisticRegression()
# Train the model on the transformed training data
model.fit(X_train_transformed, y_train)
# Predict the ratings for the testing data
y_pred = model.predict(X_test_transformed)
# Evaluate the model's performance
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
# Display the confusion matrix
print(confusion_matrix(y_test, y_pred))
```

To this:

```
pipe = pipeline("sentiment-analysis")
pipe([document_list])
```

What the Hugging Face pipeline does

- ❑ Handles all the preprocessing.
- ❑ Handles all the postprocessing.
- ❑ Can be used for CV, NLP, Multimodal, and other common use cases.
- ❑ Contains automatic GPU support.
- ❑ Built in batching functionality available.
- ❑ Can quickly swap out models and iteratively test without having to change lots of preprocessing code.

General Steps:

Input -> Tokenization -> Model Inference
-> Post-Processing (task dependent) ->
Output

Matches preprocessing steps to the model and checkpoint you are using for analysis.



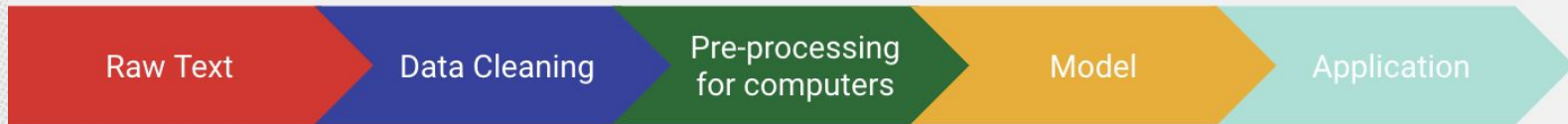
Hugging Face Pipeline Components

- Pre-processing details:
 - a tokenizer, image_processor, feature_extractor, or processor.
- **Model selection**
- (optional) Post-processing details

This single API captures most of our workflow!

You may need to still do some data cleaning (like deduplication, or deidentification) but many other steps are covered!

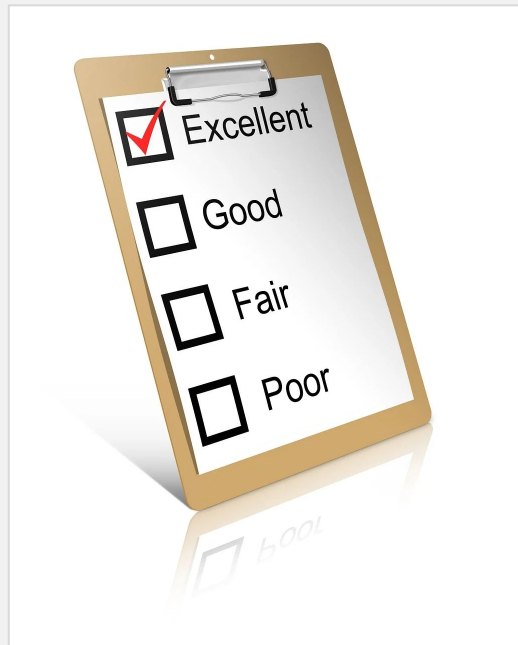
NLP General Workflow



Focus on performance, not preprocessing

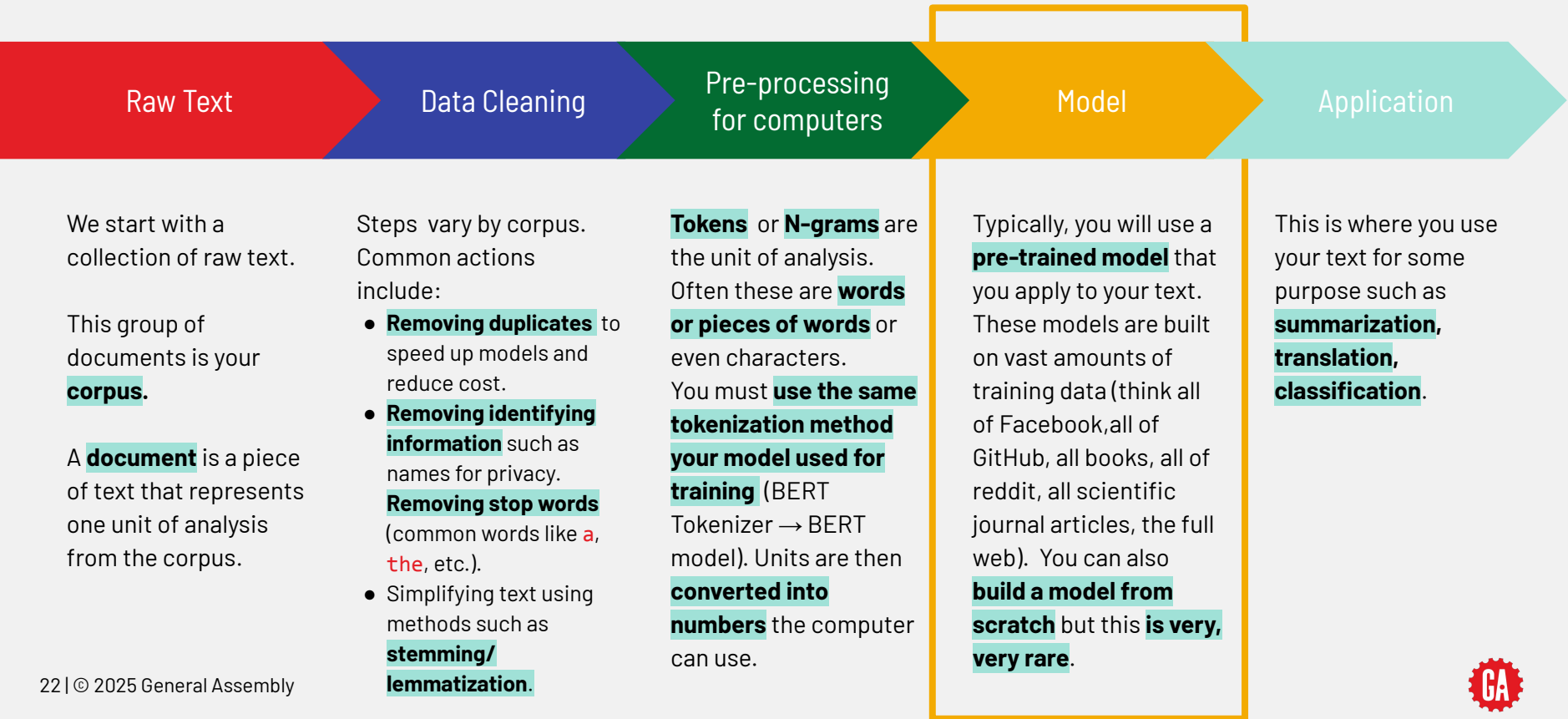
Instead of focusing the bulk of time on proper preprocessing and setup, now you focus on model selection and performance that produces the best result for your business problem.

Let's use sentiment analysis as an example.



Model Selection

NLP General Workflow

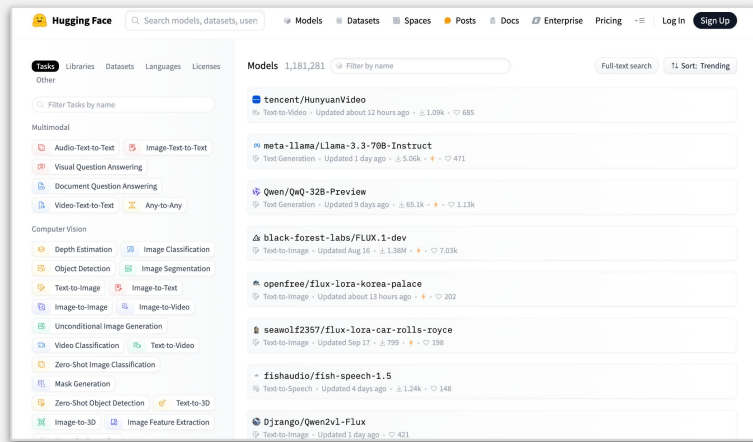


Model selection tips

The model you select should be matched to your data.

- Working with X/Twitter posts? Use a model that has seen tweets before.
- Using customer reviews? Use a model that is trained on customer reviews.

Large models can handle most public data, but may struggle with specialized data such as medical notes.



How to pick a model

1

What is your task?

(Text classification, summarisation, named entity recognition, etc.)

2

Do you need a multilingual model?

3

Is speed or accuracy more important for your use case?

4

Do you need a model fine-tuned on domain-specific data (e.g., X/Twitter, medical, legal)?

Steps for using Hugging Face models

1

Go to the Model Hub:
<https://huggingface.co/docs/hub/en/mod>

2

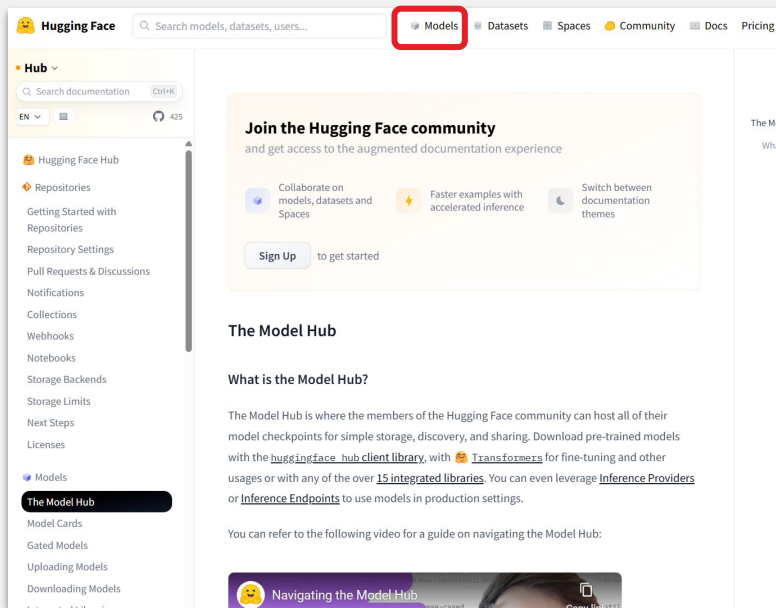
Click on the tasks to filter models for your desired task, and libraries to filter your toolkit. You can also filter by language and license type.

3

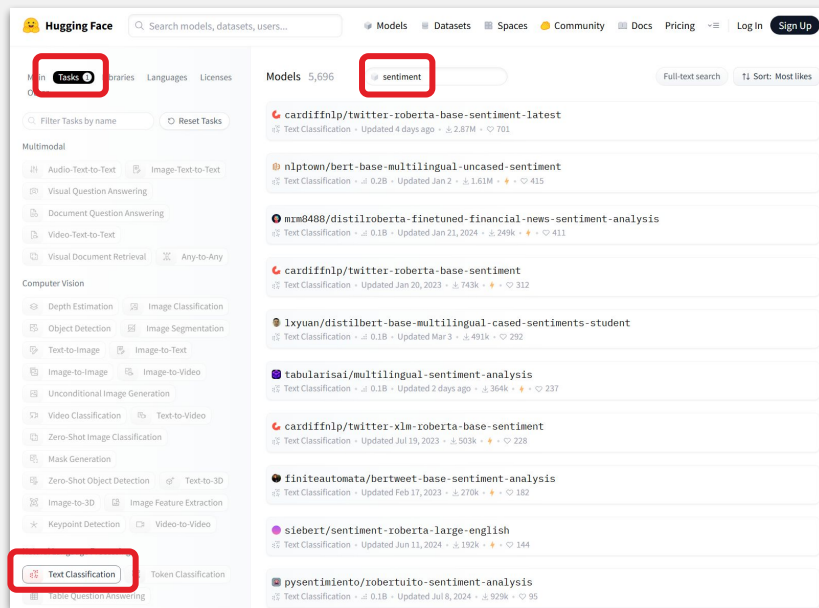
Review the model card and select a pretrained model that fits your needs.

Steps for using Hugging Face models: sentiment analysis

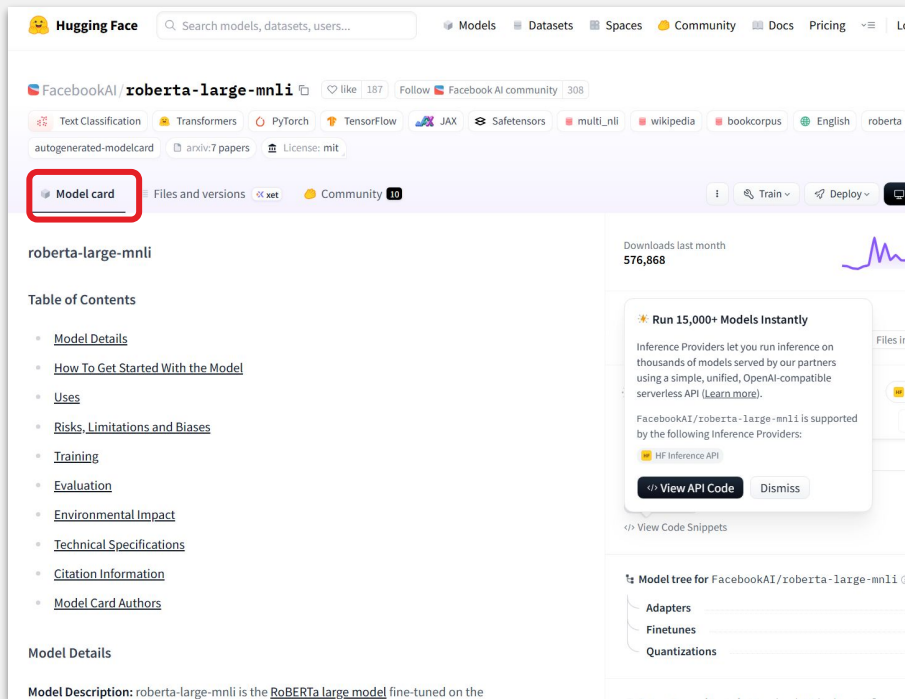
Step 1: Go to the Model Hub



Step 2: Click on the tasks (you can also search by keyword)



Steps for using Hugging Face models



Step 3: Examine the model card

Sentiment Analysis with Hugging Face Pipelines

- Open **encoder-sentiment-pipeline.ipynb**
- Use the following **encoder models** to complete different versions of sentiment analysis on customer reviews.
- Review the information for **each of the models below**:



- [distilbert-base-uncased-finetuned-sst-2-english](#)
- [FacebookAI/roberta-large-mnli](#)
- [nlptown/bert-base-multilingual-uncased-sentiment](#)

- ☐ What does each model output?
- ☐ How much time does it take to run each model?
- ☐ Which would you choose?

A note on Sentiment Analysis

Sentiment Analysis is a very helpful tool that can quickly analyse large volumes of text using a consistent scoring system.

But that scoring system isn't perfect!



Limitations of Sentiment Analysis

1

Can struggle with sarcasm, subtle, or highly domain specific text.

2

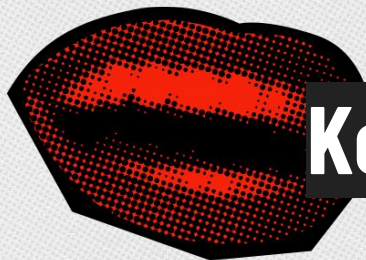
Language and cultural nuances can be missed or misinterpreted.

3

These models may be subject to deliberately misleading text.

A black and white photograph of a train platform. In the foreground, the silhouettes of three people are walking away from the camera. The platform floor is wet and reflective, showing the silhouettes of the people and the tracks in the distance. A red vertical bar is on the left side of the image.

Bring It Home



Key takeaways



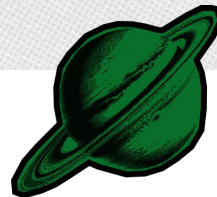
**Hugging Face Pipelines
are a wrapper for
many common tasks**

Use them for many
common language tasks.



**Select the model that
best suits your task**

Don't forget to consult the
model card and consider
compute cost.



**Match your
preprocessing to your
model**

Pipelines help manage
these steps.

DON'T FORGET:

Exit tickets



